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# Sleep Quality and Physiological Stress in Emergency Physicians During 24-Hour Shifts: A Wearable-Based Occupational Health Assessment

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**Keywords:** Circadian Rhythm; Heart Rate Variability; Sleep Architecture; Occupational Stress

**Abstract**

**Background:** *Shift-based work schedules may disrupt circadian rhythm and adversely affect sleep and stress regulation among healthcare workers. Emergency physicians are particularly exposed to high-intensity workloads and prolonged 24-hour shifts, which can increase occupational strain.* **Methods:** *This study was conducted between April and December 2024 in a tertiary emergency department. Participants wore a smartwatch continuously for 30 days to record sleep architecture and heart rate variability-based stress indicators. Physiological parameters obtained during 24-hour shifts and at home were compared. Analyses were performed across four professional-experience groups.* **Results:** *Sleep duration, deep sleep percentage, deep sleep continuity score, breathing quality score, and overall sleep score were lower during 24-hour shifts, whereas light sleep percentage, nighttime awakenings, and stress levels were higher (all $p < 0.001$). Physicians with greater professional experience had longer sleep duration and lower stress levels during 24-hour shifts compared with early-career residents. Multiple linear regression analysis indicated that professional experience was the only independent predictor of stress levels during shifts.* **Conclusions:** *Shift work in emergency medicine is associated with impaired sleep quality and increased physiological stress. Professional experience appears to mitigate stress responses but does not fully restore sleep quality. These findings support the need for targeted occupational strategies, including optimized workload distribution and structured support for early-career physicians, to improve well-being and ensure patient safety in high-intensity clinical environments.*

## 1. Introduction

The continuity of healthcare services in secondary and tertiary institutions relies heavily on shift-based and on-call duty systems. However, these demanding work patterns often conflict with the endogenous circadian rhythm, potentially precipitating chronic biological and psychological health challenges [1]. Circadian misalignment among healthcare professionals is a primary driver of sleep disturbances, which can subsequently impair cognitive functions, emotional regulation, and clinical decision-making performance [2, 3]. From an occupational safety perspective, these impairments not only increase the incidence of clinical errors but also elevate the risk of workplace accidents, thereby compromising the physical integrity of the medical staff [4]. Sleep serves as a fundamental restorative process for cognitive stability and immune homeostasis;

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consequently, sleep deprivation or fragmentation increases susceptibility to infections and occupational hazards, including needle-stick injuries and accidental exposure to hazardous biological materials [5, 6].

Beyond cognitive decline, irregular sleep-wake cycles disrupt metabolic and hormonal equilibrium, diminishing clinicians' physiological resilience [7]. Emergency departments (ED) represent a unique occupational environment characterized by high-intensity, 24/7 operations that necessitate rapid interventions under unpredictable conditions. Constant exposure to critical trauma, acute time pressure, and the necessity of managing incomplete clinical data significantly escalate occupational stress levels [8-10]. Within the framework of occupational medicine, this sustained exposure constitutes a major risk factor for physician burnout and suboptimal patient outcomes.

While the repercussions of shift work on clinician well-being are documented, the specific influence of professional experience on objectively measured sleep architecture and physiological stress responses remains inadequately characterized. Recent advancements in wearable sensor technology now enable continuous, non-invasive monitoring of these parameters in real-world clinical settings. Elucidating these physiological patterns is essential for refining occupational risk management strategies and ensuring the long-term sustainability of healthcare workforces.

In this context, exacerbated stress and impaired sleep quality among emergency physicians may jeopardize patient safety and increase the prevalence of occupational morbidity. Therefore, identifying moderating factors—such as professional seniority—is vital for the development of targeted preventive workplace interventions.

This study used smartwatch-derived metrics to assess sleep quality and stress levels among emergency physicians across different career stages. The primary objective was to determine whether professional experience influences physiological stress responses and sleep parameters during 24-hour shifts. The findings suggest that increased professional experience mitigates stress responses but does not fully restore sleep quality.

## 2. Methods

This prospective observational study was conducted between April 1 and December 1, 2024, at the Emergency Department (ED) of Manisa Celal Bayar University Hafsa Sultan Hospital. The study was designed and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. The study site is a high-acuity academic center characterized by unpredictable shift patterns and complex cognitive workloads. The primary objective was to evaluate and compare the sleep architecture, sleep quality, and physiological stress levels of emergency physicians during 24-hour on-call shifts versus off-duty periods at home, while examining the modulating role of professional experience.

A total of 13 Huawei Watch GT3 (Huawei Technologies, Shenzhen, China) smartwatches were utilized for continuous physiological monitoring. Due to the limited number of sensors, a rotation protocol was implemented; each participant underwent a continuous 30-day monitoring period before the device was reassigned to the next physician. All devices operated under identical firmware versions and application environments to eliminate potential inter-device measurement variability. This specific model was selected for its non-invasive form factor and high battery autonomy, which ensured high user compliance without interfering with clinical procedures. Data synchronization was performed via the manufacturer's health application, with automated transfer to the study database to prevent manual entry errors.

Ethical approval was granted by the Manisa Celal Bayar University Clinical Research Ethics Committee (Date: 03.04.2024; Decision No: 20.478.486//2361). The study adhered to the principles of the Declaration of Helsinki, and written informed consent was obtained from all participants prior to enrollment.

### 2.1. Smartwatch Measurement Protocol

The smartwatches provided automated analysis of sleep architecture and heart rate variability (HRV)-based stress levels. Sleep was categorized into light, deep, and rapid eye movement (REM) phases, with summary scores for deep sleep continuity and respiratory quality generated on a 0-100 scale. Stress levels, also reported on a 0-100 scale, were derived from automated HRV analysis approximately once per hour. Daily averages were utilized to reflect the autonomic nervous system response to occupational stressors and mental workload.

To ensure measurement integrity, participants were instructed to charge the devices only during daytime intervals. Nights with a wear-time of less than 90% were excluded from the final dataset. The utilization of these wearable sensors allowed for continuous, objective data collection in a real-world occupational setting, reflecting the physiological state of the participants during actual clinical practice.

### 2.2. Professional Experience Groups

Participants were categorized into four groups based on their professional experience:

- 0–1 year;
- 1–2 years;
- 2–4 years;
- specialists/academic faculty.

Residents in the first two groups primarily worked in high-volume green and yellow zones (~150-200 patients per shift). Those with 2-4 years of experience managed moderate-to-high acuity patients in the yellow and red zones (~50-100 patients per shift). Specialists and faculty physicians oversaw critically ill patients in the red zone (~15-20 patients per shift). As professional experience increased, the number of patients per 24-hour shift decreased, while case severity and clinical complexity increased. This stratification allowed for the assessment of how clinical workload and professional seniority modulate physiological stress responses in a real-world occupational setting.

The study included residents, emergency medicine specialists, and academic staff working in a shift-based system in the Emergency Department of Manisa Celal Bayar University Hafsa Sultan Hospital.

### 2.3. Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics version 26.0 (Armonk, NY, USA). Descriptive data were expressed as frequencies and percentages for categorical variables, while continuous variables were presented as mean ± standard deviation (SD), along with minimum and maximum values. The normality of data distribution was evaluated using the Kolmogorov–Smirnov test.

To compare physiological metrics obtained during 24-hour shifts and off-duty periods at home, the paired samples t-test was employed. For all paired comparisons, Cohen's d was calculated to determine the magnitude of the effect size. Differences across the four professional seniority groups were assessed using one-way ANOVA, followed by Tukey's post-hoc test for multiple comparisons where appropriate. Furthermore, univariate ANOVA models were utilized to evaluate the independent effect of professional experience on sleep and stress parameters after adjusting for potential covariates.

The relationship between continuous variables was assessed using Pearson correlation analysis. To identify independent predictors of occupational stress levels, a multiple linear regression model was constructed, incorporating variables that demonstrated significant univariate associations. To prevent multicollinearity, age was excluded from the final regression model due to its high correlation with professional experience. In all analyses, a two-tailed p-value $< 0.05$ was considered statistically significant.

## 3. Results

### 3.1. Participant Characteristics

The mean age of the participants was 29.5 ± 5.4 years, and 78 of them (78.8%) were male. Regarding professional experience, 27 participants (27.3%) were residents with 0-1 year of service, 27 (27.3%) had 1-2 years, another 27 (27.3%) had 2-4 years, and 18 participants (18.1%) were physicians with specialist or academic status.

### 3.2. Comparison of Sleep and Stress Parameters: Shift vs Home

The comparison of sleep and stress measurements obtained during 24-hour shifts and at home is summarized in Table 1. Total sleep duration, deep sleep percentage, deep sleep continuity score, breathing quality score, and overall sleep score were lower during shifts, whereas the number of nighttime awakenings, light sleep percentage, and stress levels were higher during shifts (all $p < 0.001$). No difference was observed in REM sleep percentage ($p = 0.856$).

Effect size analysis supported these findings, with Cohen's d values indicating medium to large effects for most parameters. These patterns are illustrated in Figure 1, highlighting the decline in overall sleep quality and the increase in stress levels during shift work.

### 3.3. Sex-Based Differences

No differences were observed between male and female participants in variables measured during 24-hour shifts. In off-duty measurements, men had a higher percentage of light sleep ($p = 0.012$), while women had a higher percentage of deep sleep ($p < 0.001$). No other differences were observed between male and female participants.

### 3.4. Analysis According to Professional Experience

Comparison of measurements during 24-hour shifts across professional experience groups showed differences in sleep duration, overall sleep score, and stress levels (all $p < 0.001$; Table 2). Post-hoc analyses indicated that participants with 2-4 years of experience and specialists/academic physicians had longer sleep durations and higher overall sleep scores compared to residents in the 0-1 and 1-2 year groups. Stress levels were highest in residents with 0-1 year of experience and lowest in specialist/academic physicians.

Effect size estimates for these outcomes were large (sleep duration $\eta^2p = 0.820$, overall sleep score $\eta^2p = 0.550$, and stress level $\eta^2p = 0.717$). A linear trend was observed across increasing levels of professional experience (p for trend < 0.001). No differences were observed in home-based sleep or stress parameters across professional experience groups.

### 3.5. Correlation and Regression Analyses

The correlations between physiological variables during 24-hour shifts are presented in Table 3. Stress levels were negatively correlated with age ($r = -0.595$, $p < 0.001$), overall sleep score ($r = -0.619$, $p < 0.001$), and sleep duration ($r = -0.762$, $p < 0.001$). In home-based measurements, stress level was negatively correlated with overall sleep score ($r = -0.382$, $p = 0.028$).

The results of the multiple linear regression analysis are shown in Table 4. The model was associated with stress levels ($F(3,29) = 23.18$, $p < 0.001$) and explained 67.5% of the variance (adjusted $R^2 = 0.675$). Professional experience was the only

**Table 1.** Comparison of Participants' Analyzed Values During 24-Hour Shifts and at Home.

| Variables | During 24-Hour Shifts (Mean ± SD) | At home (Mean ± SD) | p-value* | Cohen's $d_p$ |
|---|---|---|---|---|
| Sleep Duration (hours) | 4.8±1.5 | 7.7±0.8 | <0.001 | –1.81 |
| Number of Night Awakenings | 2.0±1.3 | 1.0±0.5 | <0.001 | +0.91 |
| Deep Sleep Percentage (%) | 18.4±5.6 | 28.9±4.4 | <0.001 | –1.70 |
| Deep Sleep Continuity Score | 60.7±8.6 | 76.1±6.4 | <0.001 | –1.41 |
| Light Sleep Percentage (%) | 60.3±9.4 | 49.6±5.4 | <0.001 | +1.12 |
| REM Sleep Percentage (%) | 21.3±8.6 | 21.6±4.7 | 0.856 | –0.03 |
| Breathing Quality Score | 81.8±8.3 | 89.6±6.8 | <0.001 | –1.12 |
| Overall Sleep Score | 66.6±7.9 | 79.6±4.4 | <0.001 | –1.56 |
| Stress Level Score | 65.1±20.1 | 29.3±9.3 | <0.001 | +1.33 |

** Paired-samples t-test. Effect size reported as Cohen's* $d_p$.

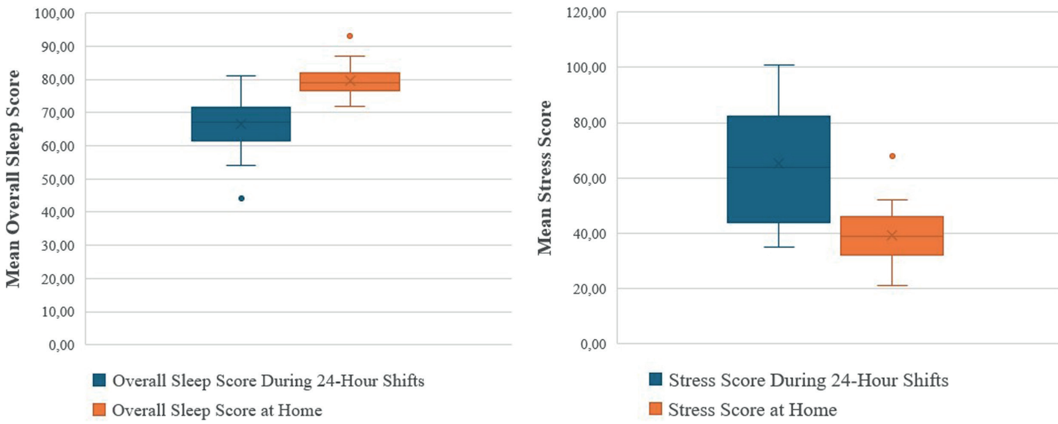


**Figure 1.** Comparison of participants' overall sleep and stress scores during work shifts and at home.

**Table 2.** Comparison of Participants' Analyzed Values During 24-Hour Shifts According to Length of Service.

| | **Length of Service** | | | | |
|---|---|---|---|---|---|
| **Variables** | **0-1 year Residents** | **1-2 years Residents** | **2-4 years Residents** | **Specialist/ Academic** | **p*** |
| Sleep Duration (hours) | 3.3±0.3[a] | 4.0±0.6[a] | 5.9±1.0[b] | 6.8±0.6[b] | <0.001 |
| Number of Nighttime Awakenings | 2.1±1.7 | 2.2±1.1 | 1.2±0.9 | 2.9±0.8 | 0.080 |
| Deep Sleep Percentage (%) | 15.3±5.9 | 18.3±7.5 | 20.5±3.0 | 20.2±2.7 | 0.191 |
| Deep Sleep Continuity Score | 56.6±8.8 | 58.9±9.9 | 64.5±6.8 | 63.8±6.5 | 0.172 |
| Light Sleep Percentage (%) | 63.0±6.8 | 61.2±13.2 | 60.3±6.0 | 54.8±10.0 | 0.416 |
| REM Sleep Percentage (%) | 21.7±5.7 | 20.6±13.4 | 19.2±5.3 | 25.1±8.3 | 0.647 |
| Breathing Quality Score | 82.3±9.1 | 81.3±10.3 | 81.6±6.9 | 82.2±8.1 | 0.995 |
| Overall Sleep Score | 60.4±7.0[a] | 62.4±5.5[a] | 71.7±4.0[b] | 74.3±5.2[b] | <0.001 |
| Stress Level Score | 88.3±9.5[a] | 67.1±13.4[b] | 56.8±13.4[b] | 40.0±3.5[c] | <0.001 |

** One-way ANOVA test was used. Different superscript letters within the same row indicate statistically significant differences ($p < 0.05$). Effect sizes for the primary outcomes were large (sleep duration $\eta^2p = 0.820$, overall sleep score $\eta^2p = 0.550$, stress level $\eta^2p = 0.717$). All significant variables demonstrated a strong linear trend across seniority levels (p for trend < 0.001).*

independent predictor of stress level (β = –0.753, p = 0.002). Sleep duration and overall sleep score during shifts were not associated with stress levels in the final model (p = 0.743 and p = 0.946, respectively).

## 4. Discussion

Sleep is a fundamental physiological need with profound metabolic, cognitive, and psychological implications, playing a pivotal role in maintaining the long-term health of the workforce [11]. While current guidelines recommend that adults obtain at least seven hours of sleep per night to maintain optimal health [12], the emergency physicians in our study averaged only 4.8 hours during 24-hour shifts. By contrast, their off-duty sleep averaged 7.7 hours, exceeding the optimal threshold. This pattern likely reflects a compensatory "rebound sleep" response, in

**Table 3.** Correlation of Participants' Analyzed Values During 24-Hour Shifts.

| Variables | Age | Sleep hours | No. of Night Awakenings | Deep Sleep Percentage (%) | Deep Sleep Continuity Score | Light Sleep Percentage (%) | REM Sleep Percentage (%) | Breathing Quality Score | Overall Sleep Score |
|---|---|---|---|---|---|---|---|---|---|
| Sleep hours | r=0.739 p<0.001 | - | - | - | - | - | - | - | - |
| No. of Night Awakenings | r=0.093 p=0.608 | r=-0.145 p=0.421 | - | - | - | - | - | - | - |
| Deep Sleep (%) | r=0.370 p=0.034 | r=0.414 p=0.017 | r=-0.484 p=0.004 | - | - | - | - | - | - |
| Deep Sleep Score | r=0.368 p=0.035 | r=0.480 p=0.005 | r=-0.492 p=0.004 | r=0.736 p<0.001 | - | - | - | - | - |
| Light Sleep (%) | r=-0.510 p=0.002 | r=-0.390 p=0.025 | r=0.068 p=0.706 | r=-0.428 p=0.013 | r=-0.288 p=0.104 | - | - | - | - |
| REM Sleep (%) | r=0.313 p=0.076 | r=0.155 p=0.389 | r=0.240 p=0.178 | r=-0.183 p=0.307 | r=-0.163 p=0.363 | r=-0.810 p<0.001 | - | - | - |
| Breathing Score | r=-0.039 p=0.828 | r=-0.042 p=0.816 | r=-0.207 p=0.248 | r=0.139 p=0.439 | r=0.020 p=0.913 | r=0.263 p=0.139 | r=-0.371 p=0.033 | - | - |
| Sleep Score | r=0.645 p<0.001 | r=0.819 p<0.001 | r=-0.384 p=0.027 | r=0.550 p<0.001 | r=0.679 p<0.001 | r=-0.410 p=0.018 | r=0.092 p=0.610 | r=0.088 p=0.627 | - |
| Stress Score | r=-0.595 p<0.001 | r=-0.762 p<0.001 | r=-0.047 p=0.796 | r=-0.328 p=0.063 | r=-0.225 p=0.209 | r=0.202 p=0.260 | r=-0.009 p=0.960 | r=-0.009 p=0.958 | r=-0.619 p<0.001 |

**Pearson correlation coefficient.*

**Table 4.** Multiple Linear Regression Analysis of Factors Associated With Stress Level During 24-Hour Shifts.

| Independent Variable | B | 95% CI for B | Standardized β | p-value |
|---|---|---|---|---|
| Professional Experience* | -13.98 | -22.14 to -5.82 | -0.753 | 0.002 |
| Sleep Duration During Shift (Hours) | -1.16 | -8.33 to 6.00 | -0.087 | 0.743 |
| Overall Sleep Score During Shift (0–100) | -0.03 | -0.95 to 0.89 | -0.012 | 0.946 |
| Constant | 105.85 | 61.36 to 150.33 | — | <0.001 |

** Dependent variable: stress level score during 24-hour shifts. The model was statistically significant ($F(3,29) = 23.18$, $p < 0.001$) and explained 67.5% of the variance in stress level (adjusted $R^2 = 0.675$). Seniority was the only independent predictor of stress level in the final model.*

which clinicians extend their sleep at home to mitigate the significant sleep debt accumulated during high-intensity clinical duties. From an occupational health perspective, sustained chronic sleep deprivation poses a significant risk to the sustainability and mental well-being of the emergency medicine workforce.

In our study, sleep duration during shifts increased with professional experience. This disparity likely reflects the distribution of occupational workload, as junior residents typically manage high-volume "green" and "yellow" zones with continuous patient flow, whereas senior physicians and specialists primarily oversee high-acuity clinical decision-making and supervision. This suggests that the opportunity for restorative sleep is limited not only by the shift system itself but also by the clinical hierarchy and the resulting volume of patient encounters.

Interestingly, home-based sleep duration did not vary across experience groups, indicating that the physiological drive for recovery sleep is a universal response to shift-work-induced exhaustion, regardless of career stage.

Consistent with prior research indicating compromised sleep quality among shift-working hospital personnel [13], our participants' global sleep scores were lower during shifts (66.6 ± 7.9) than at home (79.6 ± 4.4). Junior residents (0–2 years of experience) had the lowest sleep quality during active duty. This finding likely reflects the cumulative burden of a higher mental workload, limited clinical autonomy, and the inherent stress of early-career training. The strong positive correlation between age and sleep quality during shifts ($r = 0.645$) further supports the hypothesis that professional experience facilitates better physiological adaptation to the demands of prolonged duty cycles.

Deep sleep is essential for physical recovery, immune regulation, and cognitive consolidation; its deficiency is linked to emotional dysregulation and impaired cognitive performance [14]. As observed in intensive care nurses who exhibit highly compromised overall sleep quality [15], our study found a significant reduction in deep sleep percentage and continuity during 24-hour shifts. A critical finding of our research, however, is that although more experienced physicians obtained longer total sleep, they did not show superior deep sleep quality. This suggests that the "alert-readiness" required in the emergency department environment prevents physicians from reaching deeper, more restorative sleep stages, regardless of seniority. The unpredictable nature of acute clinical emergencies seems to keep the autonomic nervous system in a state of sustained arousal. In the long term, shift-based work schedules have been associated with persistent sleep disturbances, which may increase the risk of cardiovascular disease, metabolic disorders, and depression [16]. In addition, previous studies have reported that women may have higher baseline deep sleep proportions and a greater physiological need for restorative sleep than men [17, 18].

The most striking finding from our multiple linear regression analysis was that professional experience was the sole independent predictor of occupational stress. Although sleep duration and quality were associated with stress in univariate analyses, their effects were secondary to the protective role of experience. Greater seniority likely enhances situational awareness, decision-making efficiency, and perceived control—psychological buffers that mitigate the physiological stress response. These results underscore the need for structured mentorship and supportive supervisory models to help early-career physicians build the resilience required for shift-based care.

These findings underscore the importance of integrating experience-based risk assessment into occupational health policies. Seniority appears to function as a physiological buffer, suggesting that organizational support should be preferentially directed toward early-career physicians who may lack these adaptive mechanisms.

From a preventive medicine perspective, implementing objective fatigue-monitoring systems and tailoring shift intensity to clinical seniority may help reduce burnout and occupational accidents, thereby improving physician well-being and patient safety.

This study has several limitations that should be considered. First, the use of consumer-grade wearable sensors (Huawei Watch GT3), while highly practical for continuous monitoring in real-world settings, may introduce measurement variability compared to gold-standard polysomnography [19]. However, the non-invasive form factor of these devices was essential to ensure user compliance and prevent ergonomic interference during acute clinical procedures. Second, the single-center design may limit the generalizability of the findings to different healthcare systems or hospital tiers. Third, although we adjusted for major variables, unmeasured confounders such as caffeine and nicotine consumption, chronotype variations, and individual lifestyle habits may have influenced the physiological outcomes. Finally, the observational nature of the study precludes establishing direct causality. Despite these limitations, the use of wearable technology provided high ecological validity by capturing the objective physiological strain of emergency physicians during actual clinical practice. These findings support the feasibility of utilizing objective, real-time data to refine occupational risk management and improve the health and safety of healthcare professionals in high-intensity environments. Future multicenter studies with larger populations and advanced physiological monitoring techniques are needed to confirm these findings and further explore long-term occupational health outcomes.

## 5. Conclusion

In conclusion, our study demonstrates that 24-hour shifts in emergency medicine are directly linked to a significant decline in subjective sleep quality and a marked increase in physiological stress markers. While professional experience appears to confer psychological resilience and attenuate acute stress responses, it is insufficient to counteract the biological toll of sleep deprivation during active duty.

These findings underscore a critical vulnerability in current emergency department staffing models. To preserve the health of the workforce, particularly among early-career physicians, who are most sensitive to stress, healthcare administrators should prioritize evidence-based interventions. Such strategies must include optimized workload distribution, protected rest periods, and shift planning that accounts for seniority.

Ultimately, addressing the physiological demands of long-duration shifts is essential to sustaining high-quality emergency care. Enhancing physician well-being is not merely an occupational health goal; it is a fundamental necessity for ensuring patient safety and the long-term operational resilience of emergency healthcare systems.

**Institutional Review Board Statement:** This study was conducted in accordance with the principles of the Declaration of Helsinki. Ethical approval was obtained from the Manisa Celal Bayar University Clinical Research Ethics Committee (Approval Date: 03.04.2024; Decision No: 20.478.486//2361).

**Informed Consent Statement:** Informed consent was obtained from all subjects involved in the study.

**Declaration of Interest:** The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

**Author Contribution Statement:** E.S.G., conceptualization, data curation, formal analysis, writing – original draft; M.Y., data curation, formal analysis, supervision, writing – review and editing; H.D., investigation, data interpretation, writing – review and editing. All authors have read and approved the final manuscript.

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